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Power Point Presentation on
[Municipal Solid Waste Management]

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Solid Waste

“Solid waste” means and includes solid or semi-solid domestic waste, sanitary waste, commercial waste, institutional waste, catering and market waste and other non residential wastes, street sweepings etc. generated in the area under the local authorities and other entities. (MoEF&CC, SWM, 2016)

Any waste other than human excreta, urine & waste water, is called solid waste. (UNICEF, SLWM)



Solid Waste in Municipal Areas

Solid waste in rural areas generally includes



Kitchen Waste



House Sweeping Waste



Garden Waste



cattle dung & waste from
cattle sheds



waste from markets & shopping areas



Plastic Waste



Paper Waste

* In municipal areas most of the household contains organic wastes, with little quantity of inorganic wastes and it is completely free from toxic wastes.

Importance of Solid Waste Management in Municipal Areas

- ❑ Proper management of solid and liquid wastes is an important determinant of improved sanitation in any community.
- ❑ The goals of sanitation fail miserably when solid and liquid waste management aspect is not given proper attention to improve health and living environment of the community.
- ❑ Management of solid or liquid wastes in rural areas is much easier and profitable business than in urban areas due to the fact that it is mostly organic in nature and does not contain toxic compounds.
- ❑ Reduce environment pollution and make rural areas clean.
- ❑ Convert bio waste into energy for ensuring greater energy security at village level.
- ❑ Promote recycling and reuse of solid waste.
- ❑ Improper handling of solid waste is dangerous to health and environment. It can cause contamination of water supplies and air pollution.

4R Concept in Solid Waste

A number of waste prevention techniques are available, and they are commonly summarized as popularly known as 4Rs: reduction, reuse, recycling and recovery. To overcome the problem of solid wastes, following steps need to be taken:

- i. Wherever possible, waste reduction should be preferred.
- ii. Every effort should be made to reuse produced wastes
- iii. Recycling should be the third option for the wastes.
- iv. There are several options for recycling. Such options should be selected taking in view social and economical acceptability.
- v. Attempts should be made to recover materials or energy from waste which cannot be reduced, reused or recycled.

Solid Waste Segregation at Household Level

- Household waste should be sorted out or segregated at the source i.e. at the household level.
- This is to be done by generating awareness to sort out waste at the household level by keeping bio degradable and non bio degradable waste in separate color bins of 5 to 10 liters capacity each(e.g. green color bin for bio degradable waste and blue bin for non bio degradable waste)
- Reusable segregated non biodegradable waste may be reutilized at household level



The segregated waste should be Collet and Transport Separately for Treatment

Management of biodegradable Waste at Household Level

- ❑ Composting is an organized method of producing compost manure. Compost is particularly useful for organic matters.
- ❑ In rural areas there is higher percentage of organic matters making more applicability of composting technology.
- ❑ ***Technology options for composting of organic wastes***
 - Pile Method of composting
 - NADEP Method
 - Bangalore method
 - Indore method
 - Coimbatore Method
 - Vermi composting
 - Thermophilic Composting
 - Biogas Technology

Solid Waste Composting

- ❖ Composting is carried out in a simple manure pit or garbage pit (lined or unlined).
- ❖ In this process aerobic microorganisms oxidize organic compounds to carbon- dioxide and oxides of nitrogen and carbon from organic compounds is used as a source of energy, while nitrogen is recycled.
- ❖ In the composting process, due to exothermic reactions, temperature of mass rises. In areas/regions with higher rainfall composting in over ground heaps is advisable.
- ❖ The factors affecting the composting process are: (a) Micro-organisms; (b) Moisture, (c) Temperature and (d) Carbon/ Nitrogen (C/N) ratio

Vermicomposting

- Vermi compost is a natural compost of organic wastes through Earthworms, who take up organic wastes and after degradation and digestion, convert such wastes in the form of granules, rich in nitrogen content.
- Such vermin cast or vermin compost has good plant nutrients.
- This is a natural composting method being used in India and other countries for centuries.
- However, in recent years there has been systematic methods developed to enhance such composting by using improved methods and strains of earthworms in rural as well as urban areas in India also.

Advantages of Vermicompost

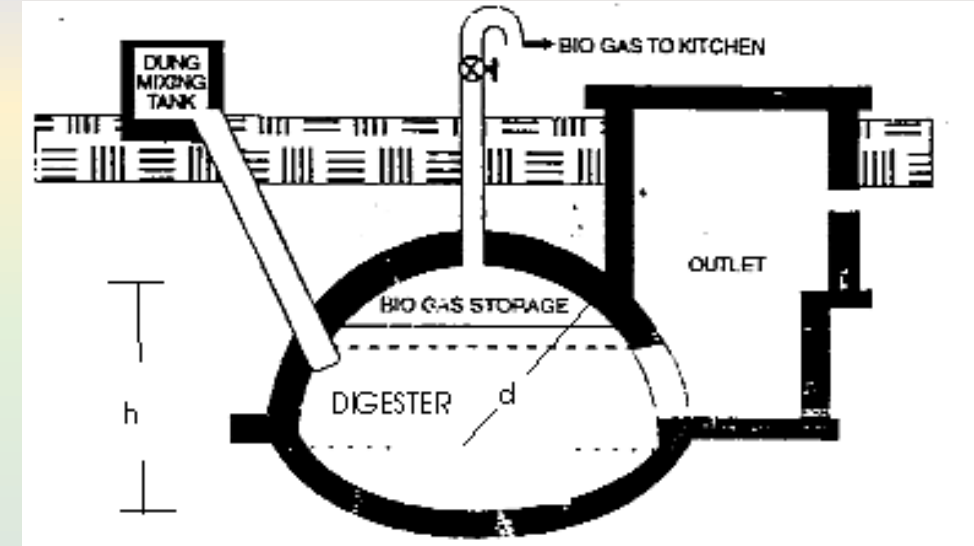
- a) Vermi compost is rich in all essential plant nutrients.
- b) Vermi compost is easy to apply, handle and store and does not have bad odour.
- c) It improves soil structure, texture, aeration, and water holding capacity. Vermi compost contains earthworm cocoon and increases the population and activity of earthworm in the soil.
- d) Vermi compost is free from pathogens, toxic elements, weed seeds etc.
- e) Vermi compost minimizes the incidence of pest and diseases.
- f) It enhances the decomposition of organic matter in soil.
- g) It contains valuable vitamins, enzymes and hormones like auxins, gibberellins etc.

Biogas Technology

- ❑ Animal dung along with human wastes can be effectively used for biogas generation through on-site biogas plant linked with toilet.
- ❑ Biogas plant is important in providing sustainable energy sources in rural communities. Generation and utilization of bioenergy has multiple advantages. It helps improve sanitation, provide bioenergy at almost nil recurring expenditure and finally slurry / effluent of biogas plant has plant nutrient value to be used for agriculture purposes.
- ❑ Thus toilet linked biogas plant has additional benefits in terms of improving sanitation.
- ❑ In villages where household wastes contain mainly organics, they are also suitable for biogas generation. Such wastes can also be mixed in the same biogas plant to generate biogas production.
- ❑ For family size biogas plant there are basically two designs (1) Floating drum type popularly known as KVIC model and (2) Fixed dome type, popularly known as Deenbandhu Model.

Deenbandhu model

- ✓ This model is predominately found in India. It is an underground fixed dome digester made up of complete brick or RCC structure.
- ✓ It is a permanent structure having almost nil operation and maintenance costs. There is no separate gas holder, biogas is stored inside plant through liquid displacement chamber.
- ✓ This design is suitable also for generation of biogas from human wastes along with cow dung and kitchen wastes.
- ✓ There is almost no effect of atmospheric variation of temperature on biogas generation during winter season.



Importance Biogas Technology

- Biogas can be used for Cooking of 3-4 family members for two times a day
- Biogas can be used for illumination with the use mantle lamps. Mantle lamps (2nos.) can be used for 6 hours per 24hrs. Such mantle lamps gives illumination equivalent to 40 watt bulb at 220 volt of electricity.
- Biogas technology also create manure which can be used in agricultural field.
- Use of Biogas helps to restrict the emission of CO₂ in atmosphere.
- It brings extra revenue to the users.

References

1. MDWS guidelines, Manuscript prepared by Dr. P.K.Jha, Consultant, Sanitation & WM, NRC, March, 2013
2. Solid and Liquid Waste Management in Rural Areas, A technical note, UNICEF
3. Guidelines on Solid and Liquid Waste Management- Asian Development Bank

Thank
you

